Portfolio Management Competition

Problem Statement, Rubrics and Resources

Each team has to prepare a portfolio consisting of stocks, bonds or commodities(silver, gold etc) present in the Indian Market. The portfolio must be tailor-made for a client whose details are mentioned in the following section. The investment plan should be structured to deliver returns that align with the client's needs over a 8-year period. Submission will be in the form of a Google Doc/Word doc downloaded as a PDF. An Excel sheet may be attached if required to support calculations or modeling.

Problem Statement

Mr. Karan Malhotra, 37, is a senior product manager at a Bengaluru-based consumer tech company. His wife, Meenal, 36, is a freelance interior designer. They have one child (age 10) and live in a rented apartment, planning to purchase their first home within the next 5–8 years.
Karan earns a post-tax salary of ₹50 lakh/year, while Meenal contributes approximately ₹17 lakh/year through project-based work. Their annual living expenses are ₹18 lakh, and they currently have no outstanding loans or EMIs. Each year, the surplus from their salaries after meeting their living expenses is invested in a non-liquid scheme, with funds locked in until the end of the 8th year, assuming zero returns.

They have a corpus amount of ₹80 lakh, which they wish to invest with a firm 8-year horizon (2025–2033), aiming to achieve the following goals:

- Buy a residential property worth ₹3.5 crore. For this, they would require ₹50 lakh in cash to pay the builder in the 5th year (2030). The remaining payment can be done anytime in the next 3 years (until the 8th year)
- Separate a sum of ₹2 crore by the 8th year to secure their child's education abroad
- Have at least ₹1.5 crore in savings after removing all these expenses to secure their future

Karan and Meenal come from financially independent families and are solely responsible for their own and their child's needs. For the analysis, no unforeseen financial withdrawals occur (medical bills, family obligations, etc). Ignore inflation in property, education or living expenses and assume no salary hikes occur.

During these 8 years, following events occur due to which you might have change your portfolio:

- May 2028: *India's lithium requirements are currently met through imports, with almost 100% import dependence. The Geological Survey of India (GSI) announces the confirmed discovery of one of the largest lithium reserves in the world in Rajasthan. Alongside, India has developed advanced extraction and manufacturing capabilities and wouldn't rely on imports for Lithium. As a result, various industries like oil and gas, battery, mining infrastructure, speciality chemicals, etc will be effected.*

- January 2031: *Indian government imposes a 30% tariff on Chinese electronics and component imports under a new Tech Sovereignty policy. This significantly disrupts supply chains across consumer electronics, telecom, semiconductors and EV components.*

Considering the client's objectives and the given scenarios, you need to construct a portfolio incorporating the necessary adjustments.

Rubrics

There is no limit to the number of pages in the document. You are encouraged to determine your own research methodology, key focus areas and the metrics you wish to emphasise. There's no single right portfolio, what matters is your reasoning, how well you support your view, and your ability to think differently. It is suggested to include the following in your report:

- **Investment Philosophy:** Explain your overall approach to investing. Are you aiming for growth, value, or capital preservation? Do you prefer a long-term, stable strategy or an active, tactical one that responds to market changes? Also mention if your portfolio follows any principles (for example; ESG) or if it avoids certain sectors and reasoning.
- **Investment Objective:** For each of your investments, clearly define the specific objective and the expected time horizon. Are you aiming to generate consistent income through dividends, accumulate funds for a major future expense, or maintain liquidity for short-term needs? Be precise about both the purpose and the timeline associated.
- **Investment Methodology:** Include a brief section for each of your investments in your report explaining how you filtered the stock/bond/commodity and the selection procedure. Mention key screening criteria, such as ratios, news, ESG/political/corporate factors or any others of your choice.

Furthermore, if time permits, after constructing your portfolio, you may calculate the following metrics for your equity segment:

1. *Beta* - Covariance/Variance
2. *Sharpe Ratio* - $(R_p - R_f)/\sigma_p$
3. *Treynor Ratio* - $(R_p - R_f)/\beta_p$
4. *Jensen Alpha* - $R_p - [R_f + (R_m - R_f)\beta]$

For calculations, use the following information:

- Use the nominal risk-free rate, fixed at 6%
- Use 5Y monthly data of stocks for variance calculation (*suggested, but you may use other time frames*)
- Market return is that of 10Y NIFTY50, ie, 10.7%

Resources and Tips

You may use tools such as ChatGPT for data extraction or concept clarification. However, all financial reasoning, selection logic, and conclusions must be your own.. While there's no need to focus heavily on the document's visual design, ensure that all content is presented in a clear, structured and professional manner.

Calculate the four specified metrics given above combined for your entire portfolio based on the weightages, only after you have fully constructed your portfolio and provided there is remaining time. To extract the 5Y monthly data of stocks:

- Investing.com

Enter stock name, get historical data after setting up the time frame, extract and download it as a CSV file and use excel functions to calculate *Beta.*

- Playlist on Financial Modelling
- Video on Sharpe, Jensen Alpha, Treynor
- Article on measuring a portfolio's performance

These playlists offer a quick overview of key terms and processes, though not in-depth. After the competition, explore the following playlist and channel for deeper understanding:

- Playlist on pricing and valuation

In case of any further doubts regarding the problem statement or guidelines, feel free to reach out to us! All the best

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